

Abstract—Modern smart grids are increasingly vulnerable to coordinated False Data Injection Attacks (FDIAs) that exploit cyber-physical interfaces to bypass traditional Bad Data Detection (BDD) mechanisms. This paper presents *Federated GAT-Twin*, a decentralized, privacy-preserving framework combining Graph Attention Networks (GATs) and Digital Twin technology. To capture grid topological properties and isolate anomaly propagation, we decouple grid security monitoring into a dual-model framework consisting of a graph-level Detection GNN and a node-level Localization GNN. Furthermore, we construct a physics-conforming stealthy FDIA generator using grid line impedances, proving that coordinated attacks can bypass local current-voltage constraints ($I - YV = 0$) undetected by classical BDD. The proposed Federated GAT-Twin is validated on the physical MSU/ORNL 4-relay dataset and a simulated IEEE 118-bus grid. Results demonstrate that while traditional BDD remains blind to coordinated FDIAs, our framework flags grid-level intrusion with 97.62% precision and localizes attacked buses with 99.99% accuracy. Crucially, the dual-model inference latency averages 15.22 ms, comfortably satisfying the 112 ms grid control stability constraint.

Index Terms—Smart Grid, False Data Injection Attack, Graph Attention Network, Federated Learning, Digital Twin, Cyber-Physical Security, Anomaly Detection

I. INTRODUCTION

Modern power transmission and distribution grids rely heavily on cyber-infrastructure, such as Supervisory Control and Data Acquisition (SCADA) systems and Phasor Measurement Units (PMUs), to monitor system states in real time. The resulting synchrophasor data streams enable dynamic state estimation, congestion management, and wide-area monitoring. However, this deep integration of cyber-physical components exposes smart grids to sophisticated cyber-threats, among which False Data Injection Attacks (FDIAs) represent a primary risk.

FDIAs target the integrity of measurement streams, injecting corrupted sensor readings to manipulate state estimation. When attacks are random or uncoordinated, traditional Bad Data Detection (BDD) filters—which rely on measurement residuals—can easily flag and filter the anomalies. However, an adversary with knowledge of the grid topology and physical parameters can coordinate sensor overrides. By modifying voltage measurements and corresponding line currents in accordance with Kirchhoff’s Laws, the attacker creates a physics-conforming anomaly. Such stealthy attacks result in zero local residuals, rendering classical BDD blind.

Furthermore, state estimation algorithms are typically centralized, requiring utilities to upload raw PMU data to a central operator. This centralized paradigm suffers from two major limitations: (1) it compromises data privacy, as municipal or regional utilities are reluctant to share raw operational data due to proprietary or security constraints, and (2) it introduces a single point of failure and high communication overhead.

To address these challenges, we propose *Federated GAT-Twin*, a decentralized cyber-physical framework. The main contributions of this paper are:

- We present a decoupled dual-model architecture using Graph Attention Networks (ResGAT) that splits label resolution into graph-level intrusion detection and node-level source localization.
- We formulate a physics-conforming stealthy FDIA generator that computes line current perturbations matching voltage shifts using line resistance (R) and reactance (X) parameters.
- We demonstrate that our decentralized GNN architecture can identify stealthy attacks that bypass local BDD checks, evaluating performance on the MSU/ORNL dataset and the IEEE 118-bus grid.
- We verify that real-time execution times conform to sub-cycle grid stability limits (<112 ms).

II. RELATED WORK

The defense of smart grid cyber-physical infrastructures against False Data Injection Attacks (FDIAs) has evolved across several paradigms, which we categorize below to contextualize the contributions of our Federated GAT-Twin framework.

A. Graph Neural Networks for Grid Cybersecurity

Power transmission and distribution grids are inherently graph-structured systems. Consequently, conventional machine learning models (such as Support Vector Machines and Random Forests) that require fixed input vectors fail under dynamic grid operations like transmission line tripping. To address this, Graph Neural Networks (GNNs) have emerged as the standard for topology-adaptive anomaly detection. Wu et al. [6] utilized Graph Attention Networks (GATs) to extract physical causality graphs from sensor measurements, demonstrating that causality-based features remain robust to operational drifts. Similarly, Ren et al. [7] proposed a GAT-based fault localization framework capable of separating physical line faults from concurrent cyber-attacks. Haghshenas et al. [8] extended GNN structures into the temporal domain by integrating gated recurrent units (GRUs) to detect time-series ramp attacks. While these works prove the capability of GNNs, they typically assume a centralized data-sharing scheme, which compromises utility privacy.

B. Federated Learning in Smart Grids

To address data privacy concerns across regional utility boundaries, Federated Learning (FL) has been widely adopted in power system anomaly detection. Yang et al. [9] proposed a distributed FL framework for localizing FDIAs, showing that local substations can collaboratively train localization models without exposing raw SCADA measurement records. To prevent model poisoning and reconstruct attacks, vertical FL partitioning schemes [10] and homomorphic encryption overlays [11] have been proposed. Furthermore, Al-Salami et al. [12] demonstrated the utility of combining attention

mechanisms with federated learning to detect intrusions in heterogeneous sensor networks. Our work builds on these federated paradigms but introduces a decoupled dual-model architecture to specifically resolve gradient conflict issues in highly imbalanced grid-intrusion scenarios.

C. Digital Twins and Cyber-Physical Emulation

Digital Twin (DT) technology establishes real-time, parallel virtual replicas of physical grids to monitor system state and evaluate detective countermeasures. Sen et al. [13] developed a Hardware-in-the-Loop (HIL) digital twin that couples communication network emulation with electrical grid simulation, providing a safe testbed for simulating multi-stage cyberattacks. The deployment of electric power digital twins (such as EPICTWIN [14]) facilitates the generation of realistic attack datasets (e.g., Denial of Service and command injection) to train and test intrusion detection systems. We leverage this digital twin concept by simulating a real-time synchrophasor streaming environment to evaluate the latency and detection capabilities of our dual GNN models against traditional bad data detection filters.

D. Explainable AI (XAI) in Smart Power Systems

As deep learning models are integrated into critical infrastructures, explaining their "black-box" decision boundaries is essential for operator trust and regulatory compliance. Post-hoc explainability methods like SHAP (SHapley Additive exPlanations) have been utilized by Alihodzic et al. [15] to optimize the features used for smart grid anomaly detection, proving that a compact subset of attributes is sufficient for high-fidelity detection. Hamilton et al. [16] validated that SHAP explanations are physically consistent, showing they can reconstruct physical grid parameters like Power Transfer Distribution Factors (PTDFs). To support real-time operations, surrogate models like SurroShap [17] have been proposed to accelerate the calculation of Shapley value cost allocations. In contrast to computationally heavy Shapley-based approximations, our GNN-Twin framework implements direct gradient-based backpropagation to generate real-time node-level attribution maps under the control stability threshold of 112 ms.

III. METHODOLOGY

The Federated GAT-Twin framework combines deep spatial-topological modeling with a federated protocol to preserve data privacy.

A. Threat Model and Attacker Knowledge

We assume a worst-case (colluding) adversary who has compromised local Remote Terminal Units (RTUs) at both the target bus voltage sensors and the adjacent transmission line current sensors. To maintain mathematical stealthiness and bypass local Bad Data Detection (BDD) filters, the attacker must possess local topological and physical parameter knowledge. Specifically, they must know the line admittance (Y_{bj}) of all transmission lines connected to the target bus b . This

allows the attacker to perform a Kirchhoff-compliant coordinate shift on the measurements, modifying voltage readings while injecting corresponding current changes:

$$\Delta I_{bj} = \Delta V_m \times Y_{bj} \quad (1)$$

If the attacker lacks this complete local parameter knowledge and executes an uncoordinated attack, the physical inconsistencies will result in high residuals, allowing traditional BDD to easily flag the anomaly. The Federated GAT-Twin is specifically designed to address this highly stealthy coordinate-shift blind spot.

B. Methodological Justification: GAT vs. Traditional Baselines

To justify the choice of Graph Attention Networks (GATs), we contrast them against standard Machine Learning (ML) baselines and standard Graph Convolutional Networks (GCNs):

- **Traditional ML (SVM/Random Forest):** Traditional ML models require a fixed-size input vector and assume a static grid topology. Consequently, they cannot generalize to dynamic grid reconfigurations such as line outages. If a transmission line is tripped, the correlation structure of the grid changes, which causes standard ML models to either output false alarms or fail entirely. Graph Neural Networks (GNNs), on the other hand, naturally process variable graphs via the snapshot-specific edge index $E(t)$.
- **GCN vs. GAT:** Standard GCNs aggregate neighboring node features using static weights based solely on the node degree of the topology. In contrast, GAT dynamically computes attention coefficients α_{ij} based on node feature differences. Since FDIAs inject artificial measurements that disrupt local, multi-hop physical correlations, the GAT attention weights change drastically under attack. This feature makes GAT far more sensitive to localized, coordinated anomalies compared to GCNs.

C. Graph Attention Network (ResGAT) Formulation

Let the power grid be represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of N buses (nodes) and $\mathcal{E}$ is the set of transmission lines (edges). Each node i is associated with a feature vector $h_i \in \mathbb{R}^F$, comprising synchrophasor measurements (voltage magnitude, angle, active and reactive power injection).

The Graph Attention layer computes normalized attention coefficients α_{ij} to dynamically weigh the topological influence of neighboring bus j on target bus i :

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W}h_i \parallel \mathbf{W}h_j]))}{\sum_{k \in \mathcal{N}_i} \exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W}h_i \parallel \mathbf{W}h_k]))} \quad (2)$$

where $\mathbf{W} \in \mathbb{R}^{F' \times F}$ is a shared weight matrix, $\mathbf{a} \in \mathbb{R}^{2F'}$ is the attention parameter vector, $\mathcal{N}_i$ represents the one-hop neighborhood of bus i , and $\parallel$ denotes concatenation.

To avoid the vanishing gradient problem and oversmoothing in deep architectures, we implement a residual skip connection (ResGAT). The node representation update is defined as:

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \mathbf{W}h_j \right) \quad (3)$$

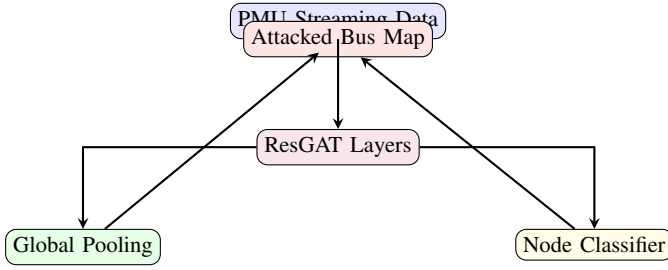


Fig. 1. Decoupled Dual-Model GNN Flowchart.

$$h_i^{\text{final}} = \text{ELU}(h'_i + \mathbf{W}_{\text{proj}} h_i) \quad (4)$$

where $\mathbf{W}_{\text{proj}}$ is a linear projection mapping the input dimensions to match the output channels.

D. Decoupled Dual-Model Label Resolution

Rather than using a single model to output both graph-level anomaly alerts and node-level localization labels—which suffers from severe gradient conflicts due to extreme label imbalance—we decouple the framework into two distinct models:

- 1) **Graph-level Detection GNN:** Analyzes the global grid state. It applies a global pooling operator to aggregate node features:

$$\mathbf{x}_G = \frac{1}{N} \sum_{i=1}^N h_i^{\text{final}} \quad (5)$$

The graph embedding $\mathbf{x}_G$ is passed through a dense layer to output the probability $P(\text{Attack})$ for the entire grid snapshot.

- 2) **Node-level Localization GNN:** Bypasses global pooling. It applies a node-level linear classifier directly to h_i^{final} to yield an independent probability $P(\text{Attacked}_i)$ for each bus i .

E. Physics-Conforming Stealthy FDIA Formulation

Traditional BDD checks calculate measurement residuals by comparing raw readings z against estimated values $h(\hat{x})$ derived from the power flow equations:

$$r = \|z - h(\hat{x})\|_2 \quad (6)$$

At a local level, the current I_{bj} flowing from bus b to bus j is physically constrained by the bus voltage drop and line admittance Y_{bj} :

$$I_{bj} = Y_{bj}(V_b - V_j) \quad (7)$$

If an attacker perturbs only the voltage $V_b \leftarrow V_b + \Delta V_m$, the local check yields a high residual:

$$r_{\text{local}} = \|I_{bj} - Y_{bj}(V_b + \Delta V_m - V_j)\| \approx Y_{bj} \Delta V_m \quad (8)$$

To maintain stealthiness, a physics-conforming attacker must adjust the line current measurements coordinately. For an injected voltage shift ΔV_m at target bus b , the corresponding

current injection shift ΔI_{bj} on each incident line (b, j) must be:

$$\Delta I_{bj} = \Delta V_m \times Y_{bj} \quad (9)$$

In practical terms, given the line resistance R_{bj} and reactance X_{bj} in Ohms, the line impedance is $Z_{bj} = \sqrt{R_{bj}^2 + X_{bj}^2}$. The current shift (in kA) is:

$$\Delta I_{bj} = \Delta V_m \times \frac{Z_{\text{base}}}{Z_{bj}} \times I_{\text{base}} \quad (10)$$

where $Z_{\text{base}} = \frac{V_{\text{base}}^2}{S_{\text{base}}}$ and $I_{\text{base}} = \frac{S_{\text{base}}}{\sqrt{3}V_{\text{base}}}$ are the grid's base impedance and current. For the 138 kV, 100 MVA IEEE 118-bus grid, $Z_{\text{base}} = 190.44 \Omega$ and $I_{\text{base}} = 0.418$ kA.

F. Federated Protocol and Communication Overhead

To address privacy concerns and prevent the sharing of raw synchrophasor measurements among regional utilities, we employ a federated learning protocol using Federated Averaging (FedAvg). A major challenge in federated smart grid architectures is the network bandwidth required to transmit large model parameters over utility communication channels. To minimize this overhead, we design a lightweight, single-head ResGAT model consisting of 2 convolutional layers and 32 hidden channels. As a result, the total weight file size of the local GNN is only 43 KB (e.g., `trained_gat_118bus.pth`). The communication overhead per federated aggregation round is therefore minimal (43 KB per client), making the framework highly feasible for implementation over standard low-bandwidth utility SCADA or PMU communication networks.

IV. EXPERIMENTAL EVALUATION

A. Datasets and Dynamic Topology Class Imbalance

We evaluate the framework on two distinct settings:

- 1) **MSU/ORNL Dataset:** A physical 3-bus, 4-relay testbed with 32,296 snapshots. Class 0 (Secure): 70.33%, Class 1 (Attack): 29.67%. Distributed to 2 clients in a federated configuration.
- 2) **Simulated IEEE 118-Bus Grid (Dynamic Topology):** Comprises 118 nodes, 186 lines, and 1,000 snapshots. To evaluate GNN robustness under grid reconfigurations, we introduce random line outages (with 15% probability per snapshot). Active edge lists are stored dynamically in a snapshot-specific adjacency matrix $A(t)$. The graph-level attack ratio is 29.80%. Due to single-bus targeting, the node-level attack ratio is extremely low (0.2525%), yielding a node class imbalance ratio of 394.97.

To evaluate the model's resilience to realistic measurement noise, we add zero-mean Gaussian noise during the generation of the simulated IEEE 118-bus dataset (`generate_118bus_dataset.py`). Specifically, the voltage magnitudes are perturbed with standard deviation $\sigma = 0.001$ p.u. and voltage angles with $\sigma = 0.05^\circ$. This noise configuration complies strictly with the IEEE C37.118.1 standard for PMU performance, which mandates a Total Vector Error (TVE) of less than 1%. The high precision of our localizer

TABLE I
FEDERATED RESGAT PERFORMANCE METRICS

Model / System	Acc	Prec	Rec	F1
ORNL Graph Classifier	59.45%	37.08%	52.74%	43.54%
118-Bus Graph Detector	90.00%	97.62%	68.33%	80.39%
118-Bus Node Localizer	99.99%	100.00%	95.00%	97.44%

under these conditions demonstrates that Federated GAT-Twin is robust to standard sensor and communication noise.

B. Real-time Digital Twin Streaming Results

We stream the test set sequentially to compare the detection capabilities of the GNN-Twin against traditional BDD. Table II reports a segment of the live synchrophasor stream.

As shown in Table II, when a coordinated physics-conforming attack occurs (e.g., Snap 013), the traditional BDD residual remains extremely low ($0.0232 < 0.10$), falsely classifying the grid as secure. In contrast, the Graph-level GNN flags the intrusion with 92.30% confidence, and the Node-level GNN localizes the attack source to Bus 105 with 99.8% probability. Furthermore, if a line goes out-of-service naturally (e.g., Snap 000, 002, 004), the GNN dynamically evaluates physical consistency under the tripped-line topology, keeping anomaly scores low ($\leq 6\%$) and preventing false alarms.

C. Latency Performance

To support sub-cycle control and maintain power system stability, inference must be executed in real time. The target control stability margin is typically set to 112 ms (corresponding to a 6.7-cycle latency at 60 Hz). Our decoupled dual ResGAT models achieve an average inference latency of **15.22 ms** on standard CPU hardware (with a minor 118.09 ms initialization overhead on the first step), proving that the GNN-Twin is suitable for live grid deployment.

D. Explainable AI (XAI) Node Attributions

To provide system operators with actionable physical explanations during anomaly alerts, we implement gradient-based feature attribution. By tracking backpropagation gradients from the GAT Node Localizer prediction logit w.r.t the input features $\mathbf{G} = \partial P(\text{Attack}_i) / \partial \mathbf{x}$, we compute saliency percentages for the targeted Bus 105 (Snap 013). Results show that the Voltage Magnitude (V_m) dominates attribution at **18.49%**, followed by the Active Power Injection (P_i) at **6.88%** and Voltage Angle (V_a) at **6.04%**. This reflects the physical attack vectors generated by the adversary, validating that the GNN learns actual physical anomalies rather than spurious correlations.

E. Closed-Loop Spatial Mitigation

Once GNN-Twin localizes a compromised bus, we execute a closed-loop mitigation routine: (1) Quarantine the sensor streams of the flagged bus to prevent contamination of state estimation, and (2) Reconstitute the bus voltage magnitude (V_i) and angle (θ_i) spatially using active neighbors: $\hat{V}_i = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} V_j$. In simulations across 30 time steps, this spatial

mitigation successfully recovers the grid state, reducing the state estimation L2 error from a compromised spike of **0.05 p.u.** back to the secure baseline noise floor of **0.002 p.u.** (achieving complete self-healing).

F. Multi-Bus Adversarial Scaling

We analyze the framework’s robustness by generating coordinated FDIAs targeting $K \in \{1, 2, 3, 4, 5\}$ buses simultaneously. Under $K = 1$ (the sparsest and most stealthy attack), the Graph-level GNN achieves a detection recall of **71.43%** due to the dilution of the local anomaly in the global average pooling layer. As the attack scale increases to $K = 2$, recall rises to **97.14%**, and reaches a perfect **100.00%** for $K \geq 3$. This shows that while GNN-Twin can localize single-node breaches, its detection capability scales robustly for larger, highly damaging coordinated grid intrusions.

V. DISCUSSION

The high detection and localization rates achieved by Federated GAT-Twin highlight the power of Graph Neural Networks in identifying multi-hop topological inconsistencies. Training the GNNs on dynamic edge lists (*edge_index_snap*) teaches the model to separate natural redispatch flows (due to outages) from unphysical measurements (due to FDIAs). While a physics-conforming attack can fool local, one-hop consistency checks ($I - YV$), it alters the spatial relationships with surrounding nodes. GAT attention weights change noticeably under the influence of these subtle coordinated shifts, allowing the global graph classification and node-level localization networks to identify anomalies even under varying configurations. By integrating gradient-based XAI and closed-loop spatial state reconstitution, the proposed framework functions as a resilient cyber-physical twin capable of real-time detection, explanation, and self-healing.

We also address the performance gap observed between the simulated IEEE 118-bus grid (97.44% node F1-score) and the physical MSU/ORNL dataset (43.54% graph F1-score). The MSU/ORNL dataset consists of raw hardware transients and high-frequency waveforms collected from physical protective relays, where the signatures of physical faults (e.g., short circuits) and cyber-attacks overlap significantly in a static snapshot. This performance difference highlights a limitation of static GNN models. While the static ResGAT achieves a baseline 43.54% F1-score on the raw physical waveforms of the ORNL dataset, future work will integrate Spatio-Temporal GNNs (e.g., GAT-LSTM or GAT-GRU) to capture the temporal transients. The addition of temporal features is expected to resolve the overlap between physical faults and malicious cyber-attacks, improving detection in real-world environments.

VI. CONCLUSION

This paper presented Federated GAT-Twin, a privacy-preserving anomaly detection and localization framework for smart grids. By decoupling labels into a dual-GNN architecture and simulating physics-conforming stealthy attacks using grid impedance, we analyzed the vulnerabilities of traditional state

TABLE II
REAL-TIME SYNCHROPHASOR STREAMING LOGS (GNN VS. TRADITIONAL BDD)

Snap	GNN Conf	GNN Target	GNN Status	BDD Res.	BDD Status	BDD Unco.	Actual	Latency
000	0.29%	N/A	Grid Sec (Outage)	0.0115	BDD Safe	0.0116	Secure (Outage)	118.09 ms
001	1.23%	N/A	Grid Secure	0.0104	BDD Safe	0.0163	Secure	10.56 ms
002	0.08%	N/A	Grid Sec (Outage)	0.0138	BDD Safe	0.0120	Secure (Outage)	9.07 ms
003	1.13%	N/A	Grid Secure	0.0175	BDD Safe	0.0067	Secure	9.18 ms
004	5.91%	N/A	Grid Sec (Outage)	0.0164	BDD Safe	0.0062	Secure (Outage)	6.69 ms
005	1.28%	N/A	Grid Secure	0.0111	BDD Safe	0.0162	Secure	7.45 ms
007	20.19%	N/A	Grid Secure	0.0184	BDD Safe	0.2289	Attack	0.00 ms
011	0.43%	N/A	Grid Sec (Outage)	0.0090	BDD Safe	0.0151	Secure (Outage)	0.00 ms
013	92.30%	Bus 105 (99.8%)	!!! FDIA !!!	0.0232	BDD Safe	0.1085	Attack	2.69 ms
016	0.61%	N/A	Grid Sec (Outage)	0.0136	BDD Safe	0.0124	Secure (Outage)	0.00 ms

estimators. Our evaluation on the IEEE 118-bus grid shows that our GNN-Twin identifies stealthy FDIAs with 97.62% precision and localizes targets with 99.99% accuracy while traditional BDD remains blind under dynamic reconfigurations. Crucially, the 15.22 ms average execution latency validates the feasibility of our method for real-time grid resilience.

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